

Product Completion with Fuzzy Association Analysis: Recommendation System Application for an Insurance Company

Sultan Ceren Öner

Data Science and Innovation
Ata Technology Platforms
İstanbul, Türkiye
cereno@atp.com.tr

Melike Demirdağ

Data Science and Innovation
Ata Technology Platforms
İstanbul, Türkiye
meliked@atp.com.tr

Ahmet Tuğrul Bayrak

Data Science and Innovation
Ata Technology Platforms
İstanbul, Türkiye
tugrulb@atp.com.tr

Olca Taner Yıldız

Computer Engineering
Özyeğin University
İstanbul, Türkiye
olcay.yildiz@ozyegin.edu.tr

Abstract—Recommendation systems have been widely accepted and have attracted significant attention both from practitioners and academicians. Utilising past behavioural patterns or indicators of customers, recommendation systems determine items or contents that would be related to a user. From this perspective, association rule mining provides extracting rules for related items according to the sequence of past behaviour of the customers. Thus, recommendation systems can be enhanced by using these 'valuable' rules to find out what kind of items customers might prefer and what could be shown that is not provided before. This study includes the following: (i) The way of association rule mining can be used to improve state-of-the-art association rule mining algorithms (ii) an ensemble of the extracted rules can be used as an alternative way for current state-of-the-art recommendation approaches providing fuzzy set theory and (iii) a case study to prove the way for implementation of fuzzy ensemble association rule mining. The case study is conducted using an insurance dataset, and results are reported for building the infrastructure of ensemble-based recommendation systems.

Index Terms—recommendation systems, association rule mining, apriori algorithm, fuzzy ensemble

I. INTRODUCTION

Association rule mining is a significant research problem in data mining and enables giving items or rules that are mostly preferred together [1]. Although there are several approaches, such as Apriori, Apriori TID, and Apriori Hybrid, in cases where the decision cannot be fully made due to the imprecise data, results cannot be determined precisely [2]. Thus, additional alternatives can be obtained by fuzzifying the data or other fuzzy numbers as appeared in problems like product completion or presenting products together [3]. Distinguishing between interval-based crisp values restricts researchers from inferring additional alternatives for the selected choice. This problem leads to incorrect decisions in practice: values close to the interval boundary are either heavily emphasised or ignored, which can cause information loss [4]. For example, fuzzy logic is used to transform numerical attributes into fuzzy attributes like "Income = High." Precise association rules use sharp partitioning to transform numerical attributes into binary attributes like "Income = [100k and above]," and these short

intervals potentially cause information loss [5, 6]. Fuzzy logic is used to convert the quantitative values of attributes into categorical ones to eliminate any information loss caused by sharp partitions. Therefore, fuzzy set theory has been accepted as a suitable tool for modelling association rules on various frequent patterns in data mining [7]. In this regard, the product completion process in recommendation systems can be made more realistic with fuzzy interval numbers, providing decision-makers with the ability to make choices.

This study discovers fuzzy association rules among operational data items to provide a more appropriate partition of an object set than clustering. As an application, a recommendation system for an insurance company's coverage selection is developed. In this coverage selection application, users can choose additional packages based on their competence and features. The aim of this study is to reveal the closest packages that match the packages selected or preferred by users using fuzzy association analysis and to ensure that the preferences of new users are formed more clearly and quickly. Since the end user's demand is accessed without intermediaries in such online platforms, being more agile and responsive to demand becomes important and enables more efficient package planning and definition activities. On the other hand, state-of-the-art approaches for association rule mining is generally extracting different rules, which can be helpful for evaluating diversified customer groups. Thus, the combination of the extracted rules is necessary to overcome this problem. To sum up, the motivation of this study is to evaluate different rules considering imprecise data and enhance more comprehensive and special rules that can not be gathered from a single association rule mining approach. This point of view provides diversity in association rule mining and more flexible choices can be extracted for the completion of the insurance packet. The contribution of interval-valued intuitionistic fuzzy numbers in the dataset is to present the data quality that implies the level of trustworthiness of the data. If the data has a higher membership value, the data converges to the crisp form of the corresponding value.

In literature, association rule mining was adapted in various

fields such as production planning [16], maritime accident evaluation [17], privacy-preserving data (stream) mining [18] and vehicle crashes [19] to determine patterns among combinations of attributes in a multivariate form that enable to realise unexpected appearances of prior conditions. Thus, the insurance packet system is a novel area for the adaptation of association rule mining to discover undetected patterns that customers had selected. In addition to that case, the authors mentioned that these combinations cannot be extracted precisely. Abu-Al Dahab et al. [20] explained the multi-labelled extraction of associated rules, yet the rules for customer relationships are gathered from evolutionary algorithms that can take much more time than expected. Besides those studies, the work of [21] emphasises that fuzzy systems for a large number of observations are still in their early stage. In [22], an overview of fuzzy techniques has been demonstrated where fuzzy approaches have been utilised to handle impreciseness in data. They also showed that most of the recent fuzzy studies only consider a limited amount of data as their definition of corresponding values.

In this study, ensemble fuzzy association rules (Fuzzy Association rule mining) are used to examine consumers' habits of requesting packages in the coverage selection application. Apriori, FP-Growth, and Eclat algorithms are used for the initial analysis, and the values that have the most influence on the selection are fuzzified to achieve a more accurate coverage selection. The results are combined considering fuzzy membership degrees and the final coverage selection is completed utilising the perspective of the combination of the extracted rules. The novelty of this study is adding interval-valued intuitionistic fuzzy numbers-based association rule mining to the literature, which represents the membership and non-membership degree as an indicator of vagueness in the dataset.

II. DATA AND PREPROCESSING

For association rule mining, we utilised the side benefit dataset, which is sampled as 953 unique users and 80 benefits. We have realised that 25 of these benefits are defined as default. Thus, we decided to use 55 of the benefits.

For further analysis, the tabular dataset is transformed as transaction data that is presented as 1 if users utilised that benefit and 0 if the user does not utilise that benefit. Some of the benefits are defined as side benefits, but their limit is 0. This implies the benefits are actually passive and we have eliminated these benefits in advance as one hot encoding. After that, minimum support, minimum and maximum combinations and minimum confidence level are specified to conduct association rule mining.

III. METHOD

A. Association Rule Analysis

Determining which products or services customers are inclined to purchase during a shopping session or consecutive shopping sessions is one way to increase sales of additional items to customers. The technique of association rule analysis

is widely used in data mining, specifically under the name of market basket analysis, to identify purchasing tendencies [8]. Association rule analysis aims to discover customers' shopping behaviours from large databases. It is used to determine which other items are purchased along with a specific item when association rules are formed. Retail stores and online platforms utilise this technique to identify their customers' purchasing patterns [9]. Various algorithms are available for exploring association rules. Apriori, Equivalence Class Transformation (Eclat), and Frequent Pattern Growth (FP-Growth) are commonly used algorithms.

B. Apriori Model

The algorithm, named after the term "prior," is based on the iteration principle. Its aim is to discover frequencies in databases. It is formed by combining K-sized sets with (K-1)-sized sets. Initially, the support ratio is calculated to create sets with one element, and the frequently occurring elements in the k-1 set form the candidate set. When the dataset is gathered, support and lift values must be calculated, and then, rules are extracted from the confidence and lift threshold. The process continues iteratively until a candidate set with frequently occurring elements cannot be formed. In this phase, operations called merging and pruning are applied to reach the result.

C. Eclat

It has entered the literature as an abbreviation of the term "Equivalence Class Transformation." It works similarly to the Apriori algorithm. In the Apriori algorithm, item-based operations are performed (e.g., items 1, 2, 3 in a shopping basket), whereas, in the ECLAT algorithm, the records in which the items occur (e.g., shopping basket 100, 200, etc.) are considered. It is suitable for smaller data sets compared to the Apriori method and is much faster than the Apriori algorithm.

D. FP-Growth

The most significant advantage of the FP-Growth Algorithm is its fast execution for large datasets and efficient utilisation of system resources. It maintains all the data in a compressed tree structure called the Frequent Pattern Tree (FP-Tree). One notable feature of this algorithm is that it only scans the database twice. In the first scan, it finds the support values for all items, and in the second scan, it constructs the tree structure. In the FP-Growth algorithm, the support value is calculated for each item, and those that fall below the user-defined threshold are eliminated. The remaining values are sorted in descending order and stored. The items with higher support values are closer to the root. It checks if an item in the transaction records exists in the tree. If it does not exist, a new node is created for that item, and its support count is incremented. If it does exist, the support count of the existing node is incremented. After the tree is built, the branches where the items occur are determined. If a single branch is identified, it represents a combination of items. If there are multiple branches, the support value is set to the

minimum support value in that branch. This way, the FP-Growth Algorithm applies a divide-and-conquer approach to divide a large itemset into smaller itemsets by following the divide-and-conquer rule. The resulting tree structure (FP-Tree) is not more significant than the original dataset [10].

The steps of the FP-Growth algorithm are as follows: 1. Calculate the minimum support value. 2. Find the frequency values (i.e., how many times they are sold). 3. Sort the items based on their importance. 4. Sort the items within the transaction based on their importance. 5. Draw the FP-Tree.

Example: Let's assume you have purchased the racing games Formula1 2019, Forza Horizon 4, and Asetta Corsa. The algorithm determines the relationship between these games and infers that the next game to be purchased will also be a racing game. It then shows racing games in advertisements to the customer. This is applicable not only to racing games but also to different data such as shopping lists and car purchase histories.

Association rules analysis calculates the relationship between products based on support and confidence criteria [11]. Support: $P(X \text{ and } Y) = \text{Number of customers who bought } X \text{ and } Y / \text{Total number of customers}$ Confidence: $P(X/Y) = P(X \text{ and } Y) / P(Y) = \text{Number of customers who bought } X \text{ and } Y / \text{Number of customers who bought } Y$ The support criterion indicates how frequently this relationship occurs in the data, while the confidence criterion indicates the likelihood of a person who purchased item Y also purchasing item X. To consider the relationship between the purchase of two items significant, both values need to be as high as possible [12]. Another criterion used for interpreting rule sets is the Lift ratio. Lift is a measure used in correlation analysis. When the calculated value is greater than 1, it indicates a positive correlation between the Antecedent and Consequent.

E. Methodology

Fuzzy association rule mining allows for an easy understanding of the dataset due to linguistic terms associated with fuzzy sets for each attribute. Many association analysis algorithms tend to generate arbitrary and dominant rules that are shown by the majority rather than producing effective rules. In almost all Association Rule Mining (ARM) algorithms, the thresholds (both confidence and support) are crisp values. This support specification may not be sufficient for queries and rule representations that require the generation of rules with linguistic terms such as "low" or "high". Fuzzy approaches deal with numerical values by mapping them to actual values, thus addressing quantitative attributes.

1) Interval-valued Fuzzy Sets and Fuzzification of Data:

In fuzzy approaches used in situations involving uncertainty and complex decision-making, different types of fuzzy sets can be utilised. Interval-valued intuitionistic fuzzy sets have been employed in various applications related to decision-making since the term is introduced by Atanassov [13]. Below are some preliminary details regarding interval-valued fuzzy sets and how they are adapted to association analysis.

Step 1. An interval-valued fuzzy set. $\tilde{A}$ in a fuzzy set defined in X , where X is a non-empty set The fuzzy number can be defined as:

$$\tilde{A} = \{ \langle x, \mu_{\tilde{A}}(x), v_{\tilde{A}}(x) \rangle; x \in X \} \quad (1)$$

Here, with $x \in X$, $\mu_{\tilde{A}}(x)$ and $v_{\tilde{A}}(x)$ represent the membership and non-membership degrees, respectively, and are represented as $\mu_{\tilde{A}}: X \rightarrow [0, 1]$ and $v_{\tilde{A}}: X \rightarrow [0, 1]$. $\mu_{\tilde{A}}(x)$ and $v_{\tilde{A}}(x)$ values must meet the following condition:

$$0 \leq \mu_{\tilde{A}}(x) + v_{\tilde{A}}(x) \leq 1 \quad (2)$$

For every intuitive fuzzy set in X ,

$$\pi_{\tilde{A}}(x) = 1 - \mu_{\tilde{A}}(x) - v_{\tilde{A}}(x) \quad (3)$$

Here, for $x \in X$, it is expected that $\pi_{\tilde{A}}(x)$, which is the degree of uncertainty whether x belongs to $\tilde{A}$ or not, satisfies the $0 \leq \pi_{\tilde{A}}(x) \leq 1$ condition.

Step 2: Arrangement of the range value intuitive range equivalents of the verbal values in the data set. These responses are given in Table I. Thus, an interval-valued fuzzy decision matrix is obtained.

TABLE I. Fuzzy Number Equivalents with Scale and Range Values for Verbal Evaluations

Corresponding Linguistic Value	Membership and non-membership degrees
Very very close	$([0.8, 1.0], [0, 0])$
Very close	$([0.7, 0.9], [0.05, 0.1])$
Close	$([0.6, 0.8], [0.1, 0.2])$
Close enough	$([0.5, 0.6], [0.2, 0.3])$
Equal	$([0.5, 0.5], [0.5, 0.5])$

The opposite value (inverse) of the fuzzy number with the interval value $\tilde{A}$, $([\mu_{ij}^-, \mu_{ij}^+], [v_{ij}^-, v_{ij}^+])$ should be written as $([v_{ij}^-, v_{ij}^+], [\mu_{ij}^-, \mu_{ij}^+])$. For example, the opposite value of the interval-valued fuzzy number, which is determined to be somewhat significant, should be written as $([0.2, 0.3], [0.5, 0.6])$.

TABLE II. Individual Performances of Methods

	No. of iterations	Frequent k-itemset	Execution time (in seconds)	No. of association rules
Apriori	2000	13	70	8655
	4000	15	91	10654
	6000	17	101	11232
	8000	17	142	12144
	10000	18	167	13123
Eclat	2000	12	81	2210
	4000	13	128	4512
	6000	14	135	6791
	8000	14	173	12013
	10000	15	181	14569
FPGrowth	2000	11	210	15643
	4000	14	388	21342
	6000	17	460	32564
	8000	19	580	36344
	10000	20	620	45934

TABLE III. Crisp Ensemble and Fuzzy Ensemble Results

Crisp ensemble	Max Voting	Weighted Average Voting	Average Voting
Frequent k-itemset	12	14	14
No. of association rules	3422	4871	4359
Fuzzy ensemble			
Frequent k-itemset	15	18	17
No. of association rules	9563	8743	7544

2) *Fuzzy association analysis with range values*: Step 3: Interval-valued fuzzy association analysis A fuzzy association rule is "If X is A, THEN Y is B" if X and Y are database feature sets and A, B contain fuzzy sets characterising X and Y, respectively. As in the binary association rule, "X is A" is called the premise of the rule, while "Y is B" is called the result of the rule [14]. Fuzzy Association Rule Algorithm works as follows: i) Candidate item sets are created as a fuzzy number with range value defined above with predefined minimum support and minimum confidence ii) k - means clustering algorithm is applied to find centroids or midpoints iii) Fuzzy sets using these centroids and the degree of membership is calculated iv) Fuzzy rules are created using the fuzzy support and confidence found.

3) *Assembling of rule association results with ranges*: Step 4: Assembling of interval-valued fuzzy result matrices. The primary purpose here is to express the outputs of different models with a single matrix. The fuzzy weighted average with range value (IIFWA) operator [15] will be used as the ensembling operator.

Let $\tilde{\alpha}_i = ([a_i, b_i], [c_i, d_i])$ ($i=1, 2, \dots, n$) be the data values generated using the interval-valued fuzzy number. Here, Q is the set of interval-valued fuzzy numbers, IIFWA: $Q_n \rightarrow Q$ is the interval-valued fuzzy weighted average coupling operator defined, and this operator is represented as follows, provided that the weight vector of α_i is $w = (w_1, w_2, \dots, w_n)$ and $\sum_{i=1}^n w_i = 1$:

$$IIFWA_w(\tilde{\alpha}_1, \tilde{\alpha}_2, \dots, \tilde{\alpha}_n) = w_1 \tilde{\alpha}_1 \oplus w_2 \tilde{\alpha}_2 \oplus \dots \oplus w_n \tilde{\alpha}_n \quad (4)$$

The expansion of this expression according to the arithmetic operators defined for the interval-valued fuzzy sets is as follows:

$$IIFWA_w(\alpha_1, \alpha_2, \dots, \alpha_n) =$$

$$\left[1 - \prod_{i=1}^n (1 - a_i)^{w_i}, 1 - \prod_{i=1}^n (1 - b_i)^{w_i} \right], \left[\left(\prod_{i=1}^n c_i \right)^{w_i}, \left(\prod_{i=1}^n d_i \right)^{w_i} \right]$$

The matrix to be obtained by using this operator is the fuzzy coupling matrix $\tilde{R}_K$ with the following interval value:

$$R_{\tilde{K}} = \begin{bmatrix} [\mu_{k_{11}}^-, \mu_{k_{11}}^+] & [\nu_{k_{11}}^-, \nu_{k_{11}}^+] & \dots & [\mu_{k_{1n}}^-, \mu_{k_{1n}}^+] & [\nu_{k_{1n}}^-, \nu_{k_{1n}}^+] \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ [\mu_{k_{n1}}^-, \mu_{k_{n1}}^+] & [\nu_{k_{n1}}^-, \nu_{k_{n1}}^+] & \dots & [\mu_{k_{nn}}^-, \mu_{k_{nn}}^+] & [\nu_{k_{nn}}^-, \nu_{k_{nn}}^+] \end{bmatrix}$$

Step 5: Calculation of score comparison matrix and interval multiplicative matrix. Calculation of score comparison matrix $\tilde{S} = (\tilde{s}_{ij})_{n \times n}$, $\tilde{S} = (\tilde{s}_{ij})_{n \times n} =$

$([\mu_{k_{ij}}^- - v_{k_{ij}}^+, \mu_{k_{ij}}^+ - v_{k_{ij}}^-])$ operation is also possible. Besides, the multiplicative interval matrix is defined as $\tilde{C} = (\tilde{c}_{ij})_{n \times n} = [10^{[\mu_{k_{ij}}^- - v_{k_{ij}}^+]}, 10^{[\mu_{k_{ij}}^+ - v_{k_{ij}}^-]}]$ and takes values ranging from 0 to 10 in this way given the score comparison matrix.

Step 6: Obtaining the priority vector. Using the multiplicative interval matrix $\tilde{C} = (\tilde{c}_{ij})_{n \times n}$, it is possible to obtain the priority vector with the following equation:

$$\tilde{w}_i = \left[\frac{\sum_{j=1}^n \tilde{c}_{ij}^-}{\sum_{i=1}^n \sum_{j=1}^n \tilde{c}_{ij}^+}, \frac{\sum_{j=1}^n \tilde{c}_{ij}^+}{\sum_{i=1}^n \sum_{j=1}^n \tilde{c}_{ij}^-} \right] = [\tilde{w}_i^-, \tilde{w}_i^+], \quad i = 1, 2, \dots, n \quad (6)$$

Step 7: Constructing the likelihood degree matrix. The weights obtained in Step 5 of the $P = (p_{ij})_{n \times n}$ likelihood matrix are compared to represent which rule is more dominant:

$$P(w_i \geq w_j) = p_{ij} = \frac{\min(L_{w_i} + L_{w_j}, \max[w_i^+ - w_j^-, 0])}{L_{w_i} + L_{w_j}} \quad (7)$$

Here it should be $p_{ij} \geq 0$, with $L_{w_i} = w_i^+ - w_i^-$ and $L_{w_j} = w_j^+ - w_j^-$ being $p_{ij} + p_{ji} = 1$.

Step 8: Prioritise the degree of likelihood. The prioritisation process, where n is the number of features, should be done as seen in equation (7):

$$w_i = \frac{1}{n} \left[\sum_{j=1}^n (p_{ij} + \frac{n}{2}) - 1 \right] \quad (8)$$

Step 9: Normalisation of the weight vector. The normalisation of the weight vector obtained in Step 7 is done with $w_i^n = \frac{w_i}{\sum_{i=1}^n w_i}$.

Step 10: All the steps from Step 3 to Step 9 are also repeated for the comparison matrices of the results for each feature, and the scores for m alternatives and n features are obtained as δ_{ij} .

Step 11: The alternative importance value P_j , which is selected by considering the importance of each attribute, is calculated by considering the equation below:

$$P_j = \sum_{i=1}^n w_i \delta_{ij}, \quad j = 1, 2, \dots, m \quad (9)$$

Step 12: The alternative with the highest importance value is selected.

Step 13: In order to measure the consistency of the model, the feature weights are changed and the results are compared.

4) *Application*: Understanding the preferences and purchasing behaviours of consumers in the retail sector is very important in terms of developing the most appropriate marketing strategies. In the study, it has been examined whether users prefer the products they prefer together for a side-benefit application platform. In addition, unlike many studies, when relationships between products are found, more detailed product groups have been studied instead of working with very

general product groups. In this way, approaches that increase customer satisfaction and loyalty can be achieved.

In the application, users can choose additional packages within specific scopes according to their authorisation and features. The aim of this study is to reveal the closest packages suitable for the packages chosen or preferred by the users and to ensure that the preferences of the new users are formed more clearly and quickly. Since the end user's demand is reached without an intermediary in such online platforms, it is essential to be more agile and sensitive to demand, and it is ensured that package planning activities are carried out in a healthier way.

In the study, first of all, the structure of transactional data is taken into account. By looking at how the data is generally distributed on the basis of the transaction, the limit values where the data are concentrated have been determined. In addition, base side benefits are eliminated and we only consider benefits that are defined as a 'packet' with an acceptable limit. Basket analysis is performed with Eclat, FP Growth and Apriori algorithms by blurring the values in this limit for each feature. In this study, fuzzy association analysis results are given for 963 individual users and 55 different products, and it is analysed what other types of products could be recommended to users on the basis of choosing the products together with different combinations. As a result, the results of these three algorithms are combined with comparison matrices.

In this study, using the Python programming language, association rule mining, which is one of the data mining techniques, and the package request habits of consumers in the application are examined. Apriori, FP Growth and Eclat algorithms are used for analysis. The initial results are based on execution time and are presented in Table II. In this table, fuzzy association rule alpha cut values are changing in order to get useful and meaningful rules. For instance, if 13 itemsets are determined and 2000 iterations run to ensemble the rules by resampling the dataset, support and confidence levels are changed as the alpha cut value and 8655 rules are generated according to this alpha cut, 0.6 which directly satisfies the confidence level.

The results obtained with the Apriori algorithm, Eclat and FP-Growth algorithm are compared. In general, the FP-Growth algorithm has computed more rules with the parameters (minimum support 0.10, confidence 0.3 and lift 3) than the Apriori algorithm, which uses the tree evolution approach to find frequently recurring item sets. Therefore, the FP-Growth algorithm is more comprehensive.

Considering this point, we need to merge all the extracted rules from these 3 algorithms in order to capture different user aspects and also enhance diversity for the side benefit recommendation system. By combining the analysis results, packages suitable for customers are defined consistently. Moreover, since the algorithm quickly and effectively detects the shopping patterns of consumers, it becomes easier to detect complex relationships in terms of preferences between products. Thus, we need to compare the results with each other. This process takes a considerable time. Thus, fuzzy-

fication of confidence level and support level provides rule overlapping and naturally, more rules can be extracted using this fuzzification process.

Fuzzification of existing methods begins with the mapping of corresponding interval fuzzy set values to the rules. For instance, if rule 2 is more relevant than Rule 3 for Rule 1, then Rule 2 will be assigned a higher fuzzy correspondence value before the implementation of k means clustering. Here, k means clustering ensures the grouping of rules based on centroids and midpoints. The following figure represents the grouping of rules and number of rules belonging to each cluster. From this figure, we have 3 main clusters and 2 sub-clusters for each main cluster for grouping all the extracted rules and the rules are distributed homogeneously. The division of main and sub-clusters are shown in Fig.1.

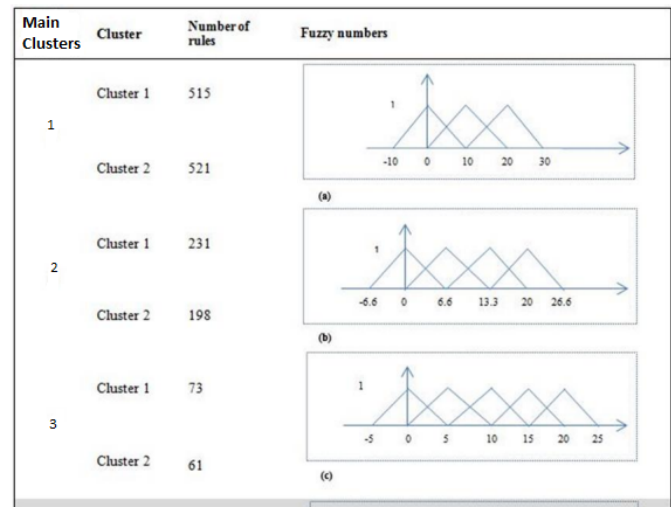


Fig. 1. Main and sub-cluster division for association rules

The interval-valued fuzzification process is used to capture the intersections between rules with the IIFWA operator for each cluster. IIFWA operator enables the combination of rules based on both relatedness and irrelevance. This distinguished approach provides certain differentiation of existing rules. Then, a fuzzy coupling matrix is constituted and a score comparison matrix is computed to represent rule centres' differentiation. In the table, main cluster 1- sub-cluster 2, main cluster 2 - sub-cluster 1 and main cluster 3- sub-cluster 1 are much more separated than the other clusters and sub-clusters. After that, the priority vector is gathered using the normalisation of the score comparison matrix. This intersection is presented in Fig.2

Finally, the likelihood degree matrix for 3 association rule approaches is obtained to represent which rule is more dominant than the others. After prioritising the degree of likelihood, the importance levels of each rule set are calculated and these rule sets are merged with the weighted sum of each confidence level and support level. An instance merged rule is set for each cluster and sub-clusters are extracted.

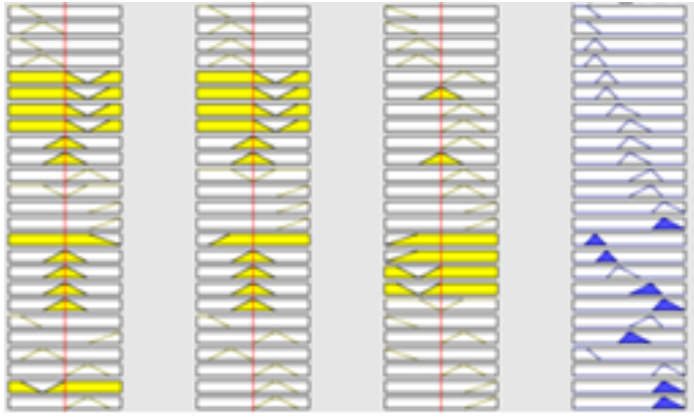


Fig. 2. Adaptation of IIFWA operator for ensemble structure

According to all these outcomes, the differentiation of each rule set is more striking than the initial k means relied on rule sets and this demonstrates the success of discrimination of rule sets which can enhance diversified rules that will be merged in the ensemble phase.

IV. COMPARISON

The combination of rules will be conducted by using ensemble methods. These methods are listed as follows:

- Max Voting: The final prediction in this technique is made based on majority voting for classification problems.
- Averaging: This technique is typically used for numerical solutions where results are average predictions. Probability can be utilised, for instance, by averaging the final classification.
- Weighted Average: Sometimes, weights are given to some models/algorithms when producing the final predictions. That approach emphasises one or two models' results.

To validate the fuzzy association rule mining-based results, we solely adapt FP growth, Eclat and Apriori separately and then merge the results that are listed above with crisp ensemble and fuzzy ensemble. The comparative table is presented in Table III:

According to this table, more frequent itemsets and, naturally, association rules are generated according to the IIFWA operator when it is utilised as a fuzzy ensemble method. This provides more diversified solutions for decision-makers and enhances diversity in association rule mining-based recommendation systems.

V. CONCLUSION

This paper includes (i) The way of association rule mining can be used to improve state-of-the-art association rule mining algorithms (ii) an ensemble of the extracted rules can be used as an alternative way for current state-of-the-art recommendation approaches providing fuzzy set theory and (iii) a case study to prove the way for implementation of fuzzy ensemble association rule mining. An application is performed for measuring the diversity of the application side

packages so that users are able to choose different packages. We concluded that diversity for recommendation systems can be provided by the adaptation of the fuzzy ensemble method with the assistance of the IIFWA operator.

For future studies, more experiments should be adapted to prove the validity of the proposed approach and different sectors could be chosen to adapt the generalisation of the method.

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